

FILTER DESIGN TEST CASE

Ladder Filter - Band 38 (2600 MHz TDD) for 5G NR Sub-6 GHz Front-End Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Ladder Filter filter targeting Band 38 (2600 MHz TDD) for 5G NR Sub-6 GHz Front-End Module. The filter is designed on AIN on Si substrate with a target center frequency of 1396.9 MHz and bandwidth of 192.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the 5G NR Sub-6 GHz Front-End Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Ladder Filter
- Frequency Band: Band 38 (2600 MHz TDD)
- Application: 5G NR Sub-6 GHz Front-End Module
- Substrate: AIN on Si
- Package: WLP (Wafer Level Package)
- Design Tool: Cadence AWR Microwave Office
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Ladder Filter design targeting Band 38 (2600 MHz TDD). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1396.9 MHz
Bandwidth (3 dB):	192.3 MHz
Insertion Loss Target:	≤ 2.4 dB
Return Loss Target:	≥ 16.0 dB
Out-of-Band Rejection:	≥ 50.0 dB
Isolation (Duplexer):	≥ 18.0 dB
Q Factor:	14492
Temperature Coefficient:	-23.9 ppm/°C
Die Size:	2.6 x 1.6 mm
Wafer Size:	6-inch
Number of Resonators:	11
Electrode Material:	Mo

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Cadence AWR Microwave Office and load the Ladder Filter template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 1396.9 MHz, BW 192.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.4 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (WLP (Wafer Level Package)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (6-inch, AlN on Si).
10. Perform on-wafer probing using Rohde & Schwarz FSW Signal Analyzer.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 4191 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.4 dB
- Return loss in passband shall be >= 16.0 dB
- Out-of-band rejection at +/- 288 MHz offset >= 50.0 dB
- Group delay variation across passband shall be <= 6 ns
- Temperature drift over -40°C to +85°C shall be <= 4.2 MHz
- Power handling shall be >= +34 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 97%

6. TEST RESULTS

Measurement Date: 2024-11-22
Devices Tested: 90
Insertion Loss (meas): 2.13 dB
Return Loss (meas): 17.6 dB
Rejection (meas): 53.3 dB
Isolation (meas): 22.5 dB
Q Factor (meas): 14492
Group Delay Variation: 2.6 ns
Power Handling: +34 dBm
Die Yield: 85.8%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

- a) The Ladder Filter design demonstrated excellent performance for Band 38 (2600 MHz TDD).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Temperature stability was within specification across full range.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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